Belief Revision in a Changing Ward

Spohn Ranks in the Epistemic State, Revision by Conditioning,
and Change No Robot Sees, on ePlanSys

Haniel Ulises

Epistemic Robotics

7 November 2026

A product update over KD45 frames cannot revise: an observation that contradicts an agent's belief leaves its accessibility relation empty, and a planner that requires consistent beliefs discards the branch. This report places belief revision inside the epistemic state an executor maintains, without changing the planner or the executor. Every world carries its Spohn rank as a grade atom; every relation remains an equivalence; belief is a formula over knowledge and the grades, and a look or a report that refutes a belief removes worlds as any observation does, after which the least grade left defines the new belief. We prove that the compilation is exact, that the revision it performs is Spohn conditioning and satisfies the AGM postulates, that an exogenous change no agent observes can be fixed at the outset with additive ranks, and that without ranks the first act admits no policy at all. On a hospital floor in Gazebo, a night porter shuts the two fire doors of a loop no robot sees him walk, and couriers that rank the nights by his habits, door by door, by full meet or not at all cross the floor to the wards, alone, in pairs that report what they find, and there and back across his second walk. Each courier plans for itself, minimising the time it expects with worlds weighted by $\varepsilon^{\text{rank}}$. Every run arrived, 144 of 144. Ranked by the porter's habits, the couriers took the least time in expectation in every act: 228.6, 238.9 and 502.1 s against 232.9, 268.2 and 527.3 s ranked door by door, and 236.8, 331.7 and 557.2 s with no belief. The difference is made on the nights a belief is refuted: on the west-loop night the explanation took 303.8 s in act I against 387.4 door by door, and 272.1 against 526.3 in act II, where the report sent the first courier into a lobby nobody had looked into. Knowledge only, which looks into lobby C on its way, was fastest on that night in act I, 211.4 s, and slowest on the rostered night, the likeliest. On the rare nights the porter's habits mislead, the explanation pays: those nights are in the tables. Without ranks, a KD45 belief admits no policy for the first act. 144 runs in Gazebo, every night of every act under all four rankings.

Contents

1 Scope **2**

2 The ward **2**

3 Ranked models **2**
3.1 Models, events and updates . . . 2
3.2 Ranks and belief . . . 3
3.3 Revision is conditioning . . . 4
3.4 A change nobody sees . . . 4
3.5 Without ranks . . . 5
3.6 Reports . . . 5

4 The domain and the four rankings **6**

5 Planning on beliefs **6**

6 The three acts **7**

7 The batch **7**

8 Implementation findings **9**

9 Limitations **10**

1 Scope

A robot that plans for what it knows waits until it knows. A robot in a building where things change unseen rarely knows anything about the parts it cannot see, and must act on what it believes: that a door shut in the evening is still shut, that the porter walked his usual round. Such beliefs are defeasible. When one is contradicted, the robot must revise it, and what it revises it to decides where it goes next.

Dynamic epistemic logic offers two treatments of belief for planning. In KD45 models, an agent's accessibility relation points at the worlds it considers possible without the truth axiom, so beliefs may be false. The product update [3] of such a model with an event that contradicts the agent's beliefs leaves its relation empty at the actual world, and the agent believes every formula. The companion report on a false belief on a warehouse floor met this as a limitation: a planner that requires consistent beliefs discards the branch, and one that does not plans on the explosion. The second treatment is plausibility models [4, 14, 10], in which each agent orders the worlds it cannot rule out and believes what holds at the most plausible of them; revision is then built into the update.

This report adds the second to an executor built for the first, without changing it. The orders are Spohn rankings [13], written into the valuation as one grade atom a world; belief is a formula over knowledge and the grades; and every relation stays an equivalence, so that plank grounds the problems, Aletheia and the planner of Section 5 plan them, and ePlanSys executes them unchanged, save for a fix to PlanSys2's executor unrelated to belief (Section 8). The demonstration is a hospital floor at night (Section 2), four rankings of the same worlds (Section 4), the analysis (Section 3), three acts in Gazebo (Sections 6 and 7), and what the implementation found (Section 8).

2 The ward

The ward is a single storey of 28 by 16 m. The south half holds the day room, the porters' lodge, the treatment room and the pharmacy, all opening onto a hall. North of the hall a band of linen rooms and sluices, 2.6 m deep, is crossed by four fire lobbies, A to D from west to east. Each runs north from the hall, turns east along a jog of 3.2 m and north again to the corridor, and its door stands across the middle of the jog, so that it is seen only from a bend inside the lobby. Four ward bays open off the corridor. The first courier starts in the pharmacy and delivers to ward 1, in the north-west; the second starts in the treatment room, between lobbies B and C, and delivers to ward 2, opposite lobby B (Figure 1).

At one in the morning the porter, a Gazebo actor moved by a plugin of the package, leaves his lodge and walks a loop: north through one lobby, along the corridor, south through another. A node swings each door shut behind him. At two, in the third act, he walks the loop the other way and opens the doors, or he is held up and walks to the day room instead. Every door is open at dusk.

Two claims about the floor carry the domain, and both are checked on the floor plan by casting rays over the occupied cells in all sixteen states of the doors: from both bends of each lobby, the rule a robot decides a door by decides it right; and from every point of the hall's and the corridor's driving lines, a quarter of a metre apart, a laser gets at most four returns off any door, fewer than the five the rule needs. Every route keeps 0.47 m from anything a robot can hit.

3 Ranked models

3.1 Models, events and updates

Definition 1 (Epistemic model [7])**.** An epistemic model over atoms P and agents $\mathcal{A}$ is $\mathcal{M} = (W, \{\sim_r\}_{r\in\mathcal{A}}, V)$ with W finite, each $\sim_r$ an equivalence and $V : W \to 2^P$. A state is a model with

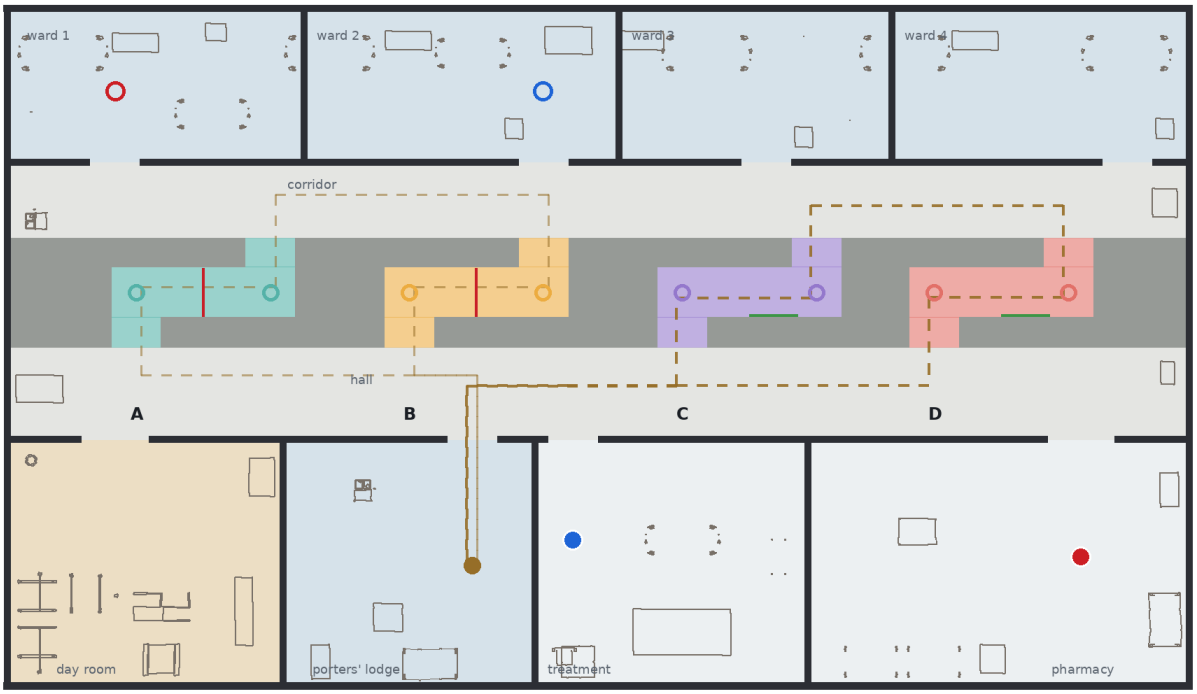


Figure 1: The ward from above, with the doors as the west loop leaves them (A and B shut), the bends a door is seen from (rings), the couriers' starts (red, blue) and beds, the porters' lodge, and in ochre the rostered loop through C and D and the west loop through A and B.

designated worlds $W_d \subseteq W$. We write $[w]_r$ for r's equivalence class of w, and $K_r\varphi$, $\hat{K}_r\varphi$ for its box and diamond.

Definition 2 (Event model and product update [3, 7])**.** An event model is $\mathcal{E} = (E, \{\sim_r^E\}, pre, post)$ with $pre(e)$ a formula and $post(e)$ assigning formulas to some atoms. The product $\mathcal{M} \otimes \mathcal{E}$ has worlds $\{(w, e) : \mathcal{M}, w \models pre(e)\}$, relations $(w, e) \sim_r (v, f)$ iff $w \sim_r v$ and $e \sim_r^E f$, and valuation $p \in V(w, e)$ iff $\mathcal{M}, w \models post(e)(p)$ where $post(e)$ defines p, and $p \in V(w)$ otherwise.

3.2 Ranks and belief

Definition 3 (Ranked model)**.** A ranked model is an epistemic model with a ranking $\kappa : W \to \mathbb{N}$. For $X \subseteq W$ let $\min_\kappa X = \{w \in X : \kappa(w) = \min_{v \in X} \kappa(v)\}$. Then $\mathcal{M}, w \models B_r\varphi$ iff $\mathcal{M}, v \models \varphi$ for every $v \in \min_\kappa [w]_r$.

This is the doxastic part of a Grove sphere system [10] or of a plausibility model [4] whose order is total within each cell, with the order given by numbers as in Spohn's ordinal conditional functions [13].

Definition 4 (Grade compilation)**.** Fix $N \geq \max \kappa$ and fresh atoms $g_0, \ldots, g_N$. The compilation of a ranked model is the epistemic model $\mathcal{M}^g$ with $V^g(w) = V(w) \cup \{g_{\kappa(w)}\}$. For φ in the language without grades, let

$$\beta_r(\varphi) = \bigvee_{k=0}^{N} \Big(K_r \textstyle\bigwedge_{j<k} \neg g_j \wedge \hat{K}_r\, g_k \wedge K_r(g_k \to \varphi) \Big), \qquad \gamma_r(\varphi) = \textstyle\bigvee_{k=0}^{N} \Big(K_r \bigwedge_{j<k} \neg g_j \wedge \hat{K}_r(g_k \wedge \varphi) \Big).$$

Proposition 1 (Belief as a formula)**.** For every world w and φ without grades, $\mathcal{M}, w \models B_r\varphi$ iff $\mathcal{M}^g, w \models \beta_r(\varphi)$, and $\mathcal{M}, w \models \neg B_r \neg\varphi$ iff $\mathcal{M}^g, w \models \gamma_r(\varphi)$. Moreover B_r satisfies KD45, K_r satisfies S5, and $K_r\varphi \to B_r\varphi$.

Proof. Let $C = [w]_r$ and $m = \min_{v \in C} \kappa(v)$; C is finite and not empty, so m exists. For $k < m$ the conjunct $\hat{K}_r g_k$ fails, since no world of C has grade k. For $k > m$ the conjunct $K_r \bigwedge_{j<k} \neg g_j$ fails, since some world of C has grade $m < k$. For $k = m$ the first two conjuncts hold, and the third says that φ holds at every world of C of grade m, which is $\min_\kappa C$. The argument for γ_r is the same with the last conjunct read as existence. $\min_\kappa C$ is not empty, which gives D; it is the same for every world of C, so $B_r\varphi$ is constant on cells, which gives 4 and 5; $\min_\kappa C \subseteq C$ gives $K_r\varphi \to B_r\varphi$. K_r is the box of an equivalence. □

3.3 Revision is conditioning

Proposition 2 (Revision is conditioning)**.** Let $\mathcal{E}$ be an event model whose postconditions define no grade atom, and set $\kappa(w, e) = \kappa(w)$. Then $(\mathcal{M} \otimes \mathcal{E})^g = \mathcal{M}^g \otimes \mathcal{E}$, and at each world of the update B_r is belief on $\min_\kappa$ of the new cell, $[(w, e)]_r = \{(v, f) : v \in [w]_r,\ f \sim_r^E e,\ \mathcal{M}, v \models \mathit{pre}(f)\}$, which is Spohn's conditioning of κ on that cell. For one agent, a public event with precondition ψ consistent with the agent's knowledge at w defines a revision $K * \psi$ of its belief set K that satisfies the AGM postulates [1].

Proof. The worlds and relations of both models are those of the product, and the grade atom of (w, e) is $g_{\kappa(w)}$ in both, since no postcondition defines it. The cell of (w, e) is as stated by the definition of the product, and Proposition 1 applied to the update gives belief on its least grade. Spohn's conditioning $\kappa(\cdot \mid X) = \kappa - \min_X \kappa$ on X has the same least worlds as κ on X. For the last claim, fix w and identify a formula with its set of worlds in $C = [w]_r$. The order $v \leq u$ iff $\kappa(v) \leq \kappa(u)$ is a total preorder on C whose least worlds are the models of K in C; it is therefore faithful to K in the sense of Katsuno and Mendelzon [11], and the operator that maps ψ to the theory of $\min_\kappa(C \cap \psi)$ is the one their representation theorem associates with it, which satisfies their postulates R1–R6 and hence the AGM postulates for the language of C. □

Looks, reports and crossings are such events here: their postconditions move robots, never grades. All four rankings of Section 4 are faithful orders, so all four couriers revise in the sense of AGM; they differ only in the order [11].

3.4 A change nobody sees

The porter's walk at two changes the doors, and every courier sees that it happens and none what it does. As an event model it has one event per outcome, all confused by every robot. Its ranks combine with the worlds' ranks additively, as for independent events in Spohn's theory and in the graded updates of Aucher [2] and van Ditmarsch [8].

Proposition 3 (A change fixed at dusk)**.** Let $\mathcal{E}$ have events e with ranks $\rho(e)$, every agent confusing all of them, every event designated. Let $\mathcal{M}^+$ have worlds (w, e) for $w \in W$, $e \in E$, valuation $V(w) \cup \{h_e\}$ for fresh atoms h_e, relations $(w, e) \sim_r (v, f)$ iff $w \sim_r v$, ranks $\kappa(w) + \rho(e)$ and designated worlds those with $w \in W_d$. Let c be the single public event with precondition $\top$ and postcondition $p := \bigvee_e (h_e \wedge \mathit{post}(e)(p))$ for every atom p some $\mathit{post}(e)$ defines. Then:

1. for any sequence of events that mention no h_e, updating $\mathcal{M}^+$ by it is, on formulas without h_e and with beliefs read by β_r, the same as updating $\mathcal{M}$ by it, worlds duplicated across E;
2. $\mathcal{M}^+ \otimes c$ is isomorphic, forgetting the h_e, to $\mathcal{M} \otimes \mathcal{E}$ with ranks $\kappa(w) + \rho(e)$.

Proof. (1) The projection $(w, e) \mapsto w$ is a bounded morphism for every $\sim_r$ and preserves atoms other than the h_e and the grades, so formulas without grades and h_e have the same truth at (w, e) as at w. For beliefs, a cell of $\mathcal{M}^+$ is $X \times E$ for a cell X of $\mathcal{M}$, and its least worlds under $\kappa + \rho$ are $\min_\kappa X \times \min_\rho E$; a formula without h_e holds on them iff it holds on $\min_\kappa X$. An event that mentions no h_e keeps this product structure, its precondition holding at (w, e) iff at w, so the claim carries along the sequence by induction. (2) The worlds of $\mathcal{M}^+ \otimes c$ are the (w, e), with the valuation $V(w)$ updated by $\mathit{post}(e)$, which is the valuation of (w, e) in $\mathcal{M} \otimes \mathcal{E}$; the relations are

$w \sim_r v$ in both, since every agent confuses all events; the ranks are $\kappa(w) + \rho(e)$ in both; and the designated worlds coincide. □

In the domain the change has two events, the porter reopening and being held up, recorded by one atom *reopens*; the grade of a world is the sum of its two walks' grades, written when the problem is written. The walk at two is a single ontic event round with effect $open_x := open_x \vee reopens$ for every door.

Proposition 4 (Open stays known, shut becomes believed)**.** Across round, $K_r\, open_x$ before implies $K_r\, open_x$ after. $K_r \neg open_x$ before implies $K_r \neg open_x$ after iff $K_r \neg reopens$; otherwise, after it, r believes the door shut iff $\neg reopens$ holds at the least-graded worlds of its cell.

Proof. round is one public event with precondition independent of the courier's cell, so cells are kept, and its effect on $open_x$ only adds truth. A world of the cell has $\neg open_x$ after the walk iff it had $\neg open_x$ and $\neg reopens$ before. The last clause is Proposition 1. □

3.5 Without ranks

Proposition 5 (Without ranks, no policy)**.** In the first act, let the courier's relation point from every world to the rostered night w_{CD} only, with the guards of Section 4 read over that relation: enter requires $\hat{K}_c\, open_x$, seek requires $\hat{K}_c$ of some door ahead open, and cross requires $open_x \wedge K_c open_x$. No policy reaches the ward in every designated world through states in which the courier's relation is serial.

Proof. Every ontic event of the domain is public with a precondition whose modal part is evaluated through the courier's relation, which points at w_{CD} from every world; such a precondition has the same truth at every world, and the update keeps every relation pointing at the copy of w_{CD}. Only a look can change that. Consider the designated world w_{AB}. To reach the ward there the courier must cross some door x with $open_x$, so $x \in \{C, D\}$; it can stand at that door's bend only after enter, which requires $\hat{K}_c\, open_x$, false for C and D at w_{CD}. It can enter only A or B, and a look there at w_{AB} observes the door shut while w_{CD} has it open, so the copy of w_{CD} is not a successor of the actual world after the look, and the courier's relation there is empty. □

Aletheia, asked for a policy with consistent beliefs, exhausts its search at depth 8 on this problem; on the graded problems it finds policies of depth 20 (explanation) and 16 (knowledge only). Without the consistency requirement the empty relation makes every belief true, and a planner may then plan on it; the companion report on the false belief describes that policy.

3.6 Reports

Proposition 6 (A report carries the belief)**.** (1) Two agents that share κ and whose cells at w coincide believe the same at w. (2) Let r's cell at w be $P \cap Q$ and s's be P, with P what is public and Q what r observed privately. After the public announcement of Q, whose precondition $K_r Q$ holds, s's cell is $P \cap Q$ and s believes what r believes. (3) If r believes φ without knowing it, the announcement of φ is not executable under a truthful precondition $K_r \varphi$; executed without it at a world where φ is false, it removes that world from s's cell, and s's knowledge is no longer truthful.

Proof. (1) Belief is determined by $\min_\kappa$ of the cell. (2) The announcement keeps the worlds of Q, so s's cell becomes $P \cap Q$, and (1) applies. (3) A public announcement keeps only the worlds where its precondition holds; if the actual world is not among them, the actual world is not in s's new cell, so s knows a falsehood. □

The domain reports only doors found shut, with precondition $K_r \neg open_x$. A courier also learns from what its teammate visibly does: a crossing's precondition tells everyone the door is open, and an entry tells everyone that the teammate does not believe the door shut, which over the grades says something about what the teammate saw.

ranking	CD	AB	AC	AD	BC	BD	at two	
explanation	0	1	2	2	2	2	+ 0, − 1	the porter's habits
door by door	0	4	2	2	2	2	+ 2, − 0	Hamming distance [6]
full meet	0	1	1	1	1	1	+ 1, − 0	capped at 1
knowledge only	0	0	0	0	0	0	0	

Table 1: The grades of the nights: for the loop at one, and added for the walk at two in act III. Door by door ranks a night by how many doors differ from what the courier expects, the rostered loop at one and no change at two; full meet believes the rostered picture and ranks everything else alike.

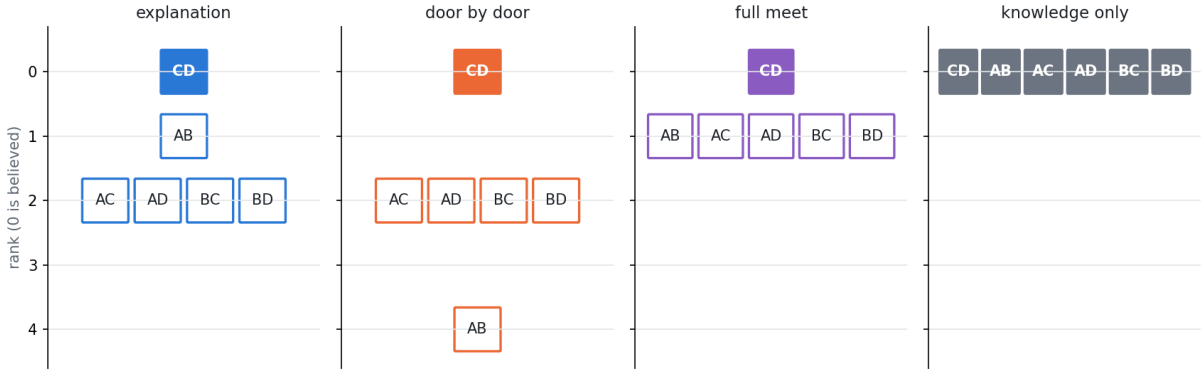


Figure 2: The six loops by rank under each ranking; the believed loops filled.

4 The domain and the four rankings

The EPDDL domain is written by `tools/model.py`. Its atoms are the doors' states $open_a, \ldots, open_d$, the grades $g_0, \ldots, g_6$, each robot's place, its leg out or back, and *reopens*. Its actions, for a courier r:

- seek: a hop along the hall, or along the corridor on the way back, guarded by γ_r of some door ahead open;
- enter the lobby of door x to the bend, guarded by $\gamma_r(open_x)$; back out to the mouth;
- look at x from the bend: semi-private sensing, r fully, the others partially;
- cross through x to the far mouth, requiring $open_x \wedge K_r\, open_x$;
- walk on the far side, deliver, leave and arrive, unguarded;
- tell x: a public announcement with precondition $\neg open_x \wedge K_r \neg open_x$;
- round, the porter's walk at two, once both couriers are in their wards.

The worlds of acts I and II are the six loops $\{x, y\}$ of two doors; those of act III are the loops crossed with reopening (+) or being held up (−). Each problem gives every courier the universal relation over its worlds and labels each world with its grade under one of the four rankings (Table 1). The grades are the only difference between the four problems of an act.

The executor's vocabulary maps every grounded action to one of five classical actions, drive, lobby, look, tell and round, whose facts exist only for ePlanSys's parallel groups: a lobby action holds its lobby, so two robots are never dispatched into one together, and a report and the porter's round write a fact every action reads, so each goes alone.

5 Planning on beliefs

Each courier chooses from what it knows and how it ranks the worlds it cannot rule out, and from nothing else. Its cell is read off the product update of the grounded task along the policy, so it holds what the courier saw, what it heard and what it saw its teammate do. Its objective is the time it expects to take to the end of its mission, with each world of its cell weighted $\varepsilon^{\kappa(w)}$, $\varepsilon = 1/8$; a courier with no beliefs minimises its worst time, the sum breaking ties. The objective is additive

over the branches of a look, so a dynamic programme over the courier's own states (place, leg, cell) finds the optimum, with the guards of Section 4 restricting its choices.

Proposition 7 (Orders of magnitude)**.** For a finite set of policies and times $T_\pi(w)$, as $\varepsilon \to 0$ the policies minimising $\sum_w \varepsilon^{\kappa(w)} T_\pi(w)$ are eventually the lexicographic minimisers of $\big(\sum_{\kappa(w)=0} T_\pi(w), \sum_{\kappa(w)=1} T_\pi(w), \dots$ If the frequencies of the worlds are proportional to ε^κ, the objective is the expected time up to a constant.

Proof. The difference of two policies' objectives is a polynomial in ε whose lowest-order nonzero coefficient is the first nonzero component of the difference of their vectors; for small ε its sign is that coefficient's. The second claim is immediate. □

The limit is not what a robot should use. On the floor first built, a look into lobby C on the way to B cost four seconds; it is worth nothing on the rostered night and much on the others, and the lexicographic believer never took it and lost to knowledge only, which did. With $\varepsilon = 1/8$ the weights are the porter's frequencies in this report, 0.842 for the rostered loop, 0.105 for the west loop and 0.013 for each other, and 0.889 for reopening at two, and for one courier the policy is optimal in expectation under them.

A team is two couriers each planning for itself. A courier that finds a door shut reports it while its teammate is still on its way. The team's policy is one tree, built in the order ePlanSys dispatches it: at each step every free courier names its next action, actions of different couriers are dispatched together, a look ends its group because it branches, and a report or the porter's round goes alone. The tree is written in Aletheia's policy format by `tools/plan.py`, which the AletheiaPlanSolver plugin calls as its planner; the plugin validates the policy against the grounded task before the executor runs it, and `tools/policy_check.py` replays every policy through an independent implementation of the product update and checks, at every node, that each courier's belief read off the grades is the one the ranks give: 5 155 beliefs over the twelve policies, with no disagreement.

6 The three acts

Act I, one courier. From the pharmacy to ward 1. The three rankings that believe the rostered loop head for B, the nearest door believed open; knowledge only enters C on its way. On the west-loop night B is shut. Ranked by explanation, the courier believes A shut too and turns back to C; door by door, it believes A open, drives to it, finds it shut, and only then goes to C; full meet goes to A as well.

Act II, two couriers. The second courier, starting between B and C, probes B first and reports it shut. The first, still in the hall, revises as it would on seeing it: ranked by explanation, it enters lobby C, into which nobody has looked, and the second learns that C is open without looking by seeing the first cross it. Door by door, the same report leaves both believing A open.

Act III, there and back. Both deliver, wait while the porter walks at two, and return. By Proposition 4, a door seen open on the way out is still known open; a door seen shut is now believed shut or reopened by the grades. Ranked by explanation the second courier returns through B, which it saw shut; on a night the porter was held up it finds B shut again, and revises again.

7 The batch

Every night of every act was run under all four rankings in Gazebo 11, the porter walking and the doors swinging, from the same starts perturbed by up to 5 cm and 5°: in acts I and II each of the six nights twice, in act III each of the twelve once, 144 runs on seven lanes. A run's time is measured from the start of its first action to the end of its last, in simulated seconds; act III's

act	ranking	night	the doors looked at, in order	predicted
I	explanation	AB	c: B shut; c: C open	361
I	door by door	AB	c: B shut; c: A shut	467
I	full meet	AB	c: B shut; c: A shut	467
I	knowledge only	AB	c: C open	248
II	explanation	AB	d: B shut; c: C open	321
II	door by door	AB	d: B shut; c: A shut	519
II	full meet	AB	d: B shut; c: C open	321
II	knowledge only	AB	d: C open	309
III	explanation	AB+	d: B shut; c: C open; d: B open (back)	571
III	explanation	AB−	d: B shut; c: C open; d: B shut (back)	625
III	door by door	AB+	d: B shut; c: A shut	785
III	door by door	AB−	d: B shut; c: A shut	785
III	full meet	AB+	d: B shut; c: C open	580
III	full meet	AB−	d: B shut; c: C open	580
III	knowledge only	AB+	c: C open; d: D open	530
III	knowledge only	AB−	c: C open; d: D open	530

Table 2: The west-loop night under each ranking's policy: every look in order, by the courier (c) or the second courier (d), with what it found, and the time `tools/plan.py` predicts.

act	ranking	expected	predicted	west loop	arrived
I	explanation	**228.6**	263.3	303.8	12/12
	door by door	232.9	272.0	387.4	12/12
	full meet	231.7	272.0	388.1	12/12
	knowledge only	236.8	280.5	211.4	12/12
II	explanation	**238.9**	267.2	272.1	12/12
	door by door	268.2	287.3	526.3	12/12
	full meet	242.3	267.2	270.6	12/12
	knowledge only	331.7	364.5	273.7	12/12
III	explanation	**502.1**	518.9	527.1 / 540.2	12/12
	door by door	527.3	540.9	793.2 / 778.3	12/12
	full meet	504.8	519.2	547.2 / 536.6	12/12
	knowledge only	557.2	570.9	535.9 / 532.4	12/12

Table 3: Expected mission time in simulated seconds, measured and predicted, under the porter's frequencies; the mean on the west-loop night (act III: reopened / held up); and the runs that arrived. In bold, the least expected time of each act.

includes the porter's walk at two, the same for every ranking. Expectations weight each night by the frequencies of Section 5.

In act I the expected times were 228.6 s explanation, 231.7 s full meet, 232.9 s door by door, 236.8 s knowledge only. Where its policy differs from another ranking's, the explanation was slower by more than 10 s on AB, by 92 s against knowledge only (frequency 0.105); BC, by 86 s against door by door (frequency 0.013); BD, by 90 s against knowledge only (frequency 0.013). In act II the expected times were 238.9 s explanation, 242.3 s full meet, 268.2 s door by door, 331.7 s knowledge only. Where its policy differs from another ranking's, the explanation was slower by more than 10 s on BC, by 50 s against knowledge only (frequency 0.013). In act III the expected times were 502.1 s explanation, 504.8 s full meet, 527.3 s door by door, 557.2 s knowledge only. Where its policy differs from another ranking's, the explanation was slower by more than 10 s on BC-, by 32 s against door by door (frequency 0.001); BD-, by 21 s against full meet (frequency 0.001). The predicted times were longer than the measured by 27.9 s on average: the planner's speed of 0.17 m/s is conservative, and the executor overlaps some actions the prediction runs apart.

In act I the explanation's policy had the least expected time for every share of west-loop nights from 0.025 to 0.20, and knowledge only from 0.25: when the "rare" loop is that common, its rank is wrong; it matched the policy planned on the frequencies themselves, the optimum for one robot,

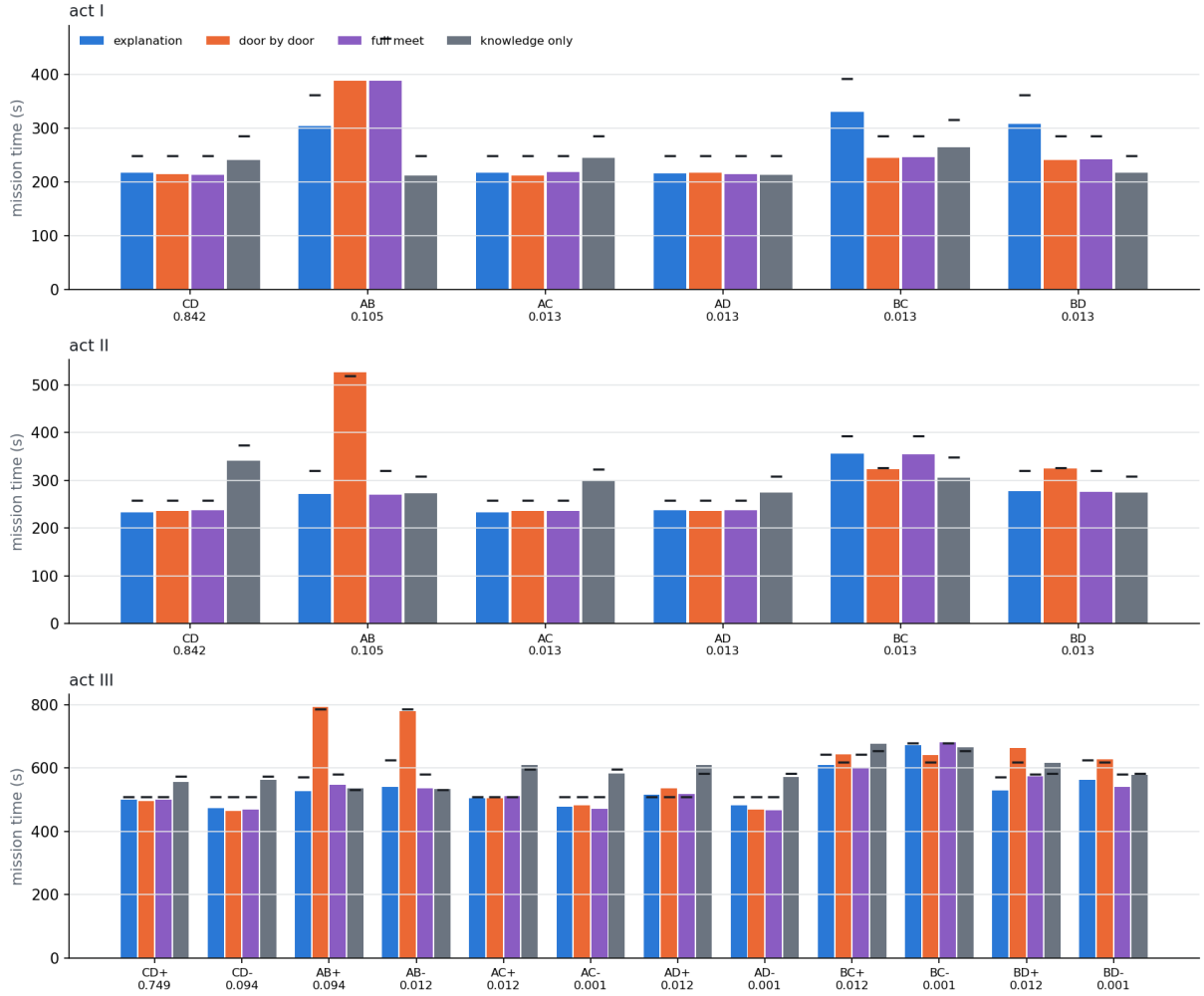


Figure 3: Mission time per night and ranking, the mean of its runs; the tick is the time `tools/plan.py` predicted for the same policy and night. Under each night its frequency.

from 0.025 to 0.20. In act II the explanation's policy had the least expected time for every share of west-loop nights from 0.025 to 0.40. In act III the explanation's policy had the least expected time for every share of west-loop nights from 0.000 to 0.40.

8 Implementation findings

A lexicographic believer never looks. Proposition 7 and the paragraph after it: the first objective, lexicographic in the ranks, passed a cheap look, and the floor and the objective were both changed.

A visible action is a message. The first team planner tracked what a courier knew from its own looks and the reports it heard. In the grounded task a courier also sees its teammate act and learns each action's precondition. The policies were valid, since the guards were checked against the true update, but held looks the courier did not need; the planner now reads each courier's cell off the product update itself.

The executor's schedule is not the policy's order. ePlanSys 0.3 moves a branching look ahead of an independent drive and copies the drive into both branches, to dispatch more together. What each robot does is unchanged; the time predicted with plain parallel groups differs from the measured by 31.8 s (9 %) on average.

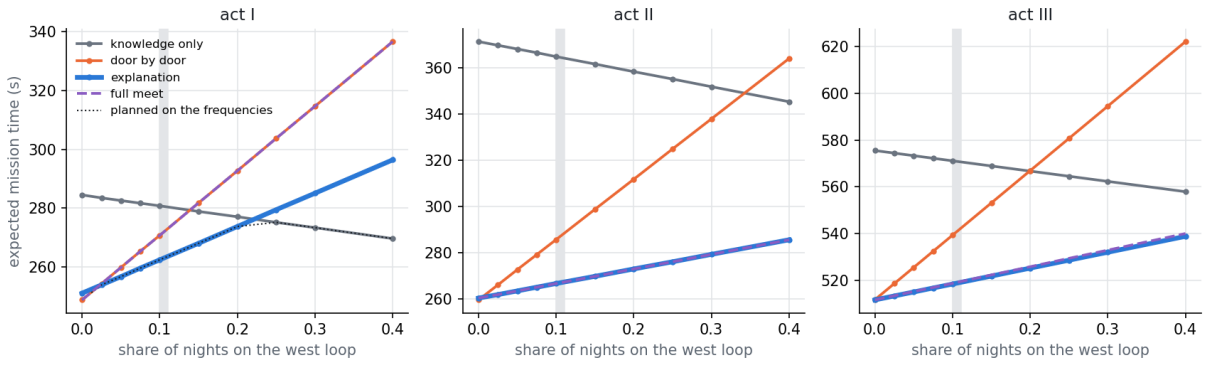


Figure 4: Each ranking's policy, planned once from its ranks, charged its predicted time under other porter frequencies: the west loop on the share of nights on the axis, the four rare loops on 5% between them, the rostered loop on the rest. The grey band is the stated share. In act I the dotted line is the policy planned on the frequencies themselves, the optimum for one robot.

A stale request starves a performer. PlanSys2's executor, as ePlanSys 0.3 ships it, creates an action executor when a behaviour-tree node first ticks and requests a performer every second until one answers. Its `ExecuteAction::halt()` does nothing, so a node ticked once and then left keeps requesting for the rest of the mission. A performer that answers it is either confirmed for an action already done, a look waiting at the wrong lobby until its 30 s run out, or rejected a moment after the request actually due has passed it by; either way that request times out after 30 s and the policy fails. Under the load of seven lanes, 6 of 52 runs of a first batch failed so, and none in the tests run one or two at a time. A patch to the executor, distributed with the package, makes an executor that has not been ticked for three seconds stop asking. The batch reported here ran entirely with it, and no executor of it had to stop requesting.

Two couriers in one hall. The policies keep two robots out of one lobby, not out of each other's way in the hall. On the night the porter walked B-C, in act II, the courier backing out of lobby C met the second courier driving past its mouth towards D; one gave way and the other backed off, about 40 s under each ranking that sent them there. Worse, two couriers driving the same way end their drives together under the executor's groups, and the one behind waited on a place the one ahead would not leave until the drive behind had ended: one run of a first batch stood for over ten minutes and hit the time limit. A drive to a place another robot stands on now ends beside it, and acts II and III were run again with that drive. The policy does not see these costs; the batch does.

Smaller faults. The drive nodes died on start, reading a parameter array in a range-for whose temporary dies before the loop body runs; and the second courier, first placed against an instrument cart, stalled backing out of it.

9 Limitations

Only hard information is modelled. A report one does not fully trust calls for an update that changes ranks, a radical or a conservative upgrade [14]; here it would take actions that rewrite the grade atoms, and none does. The ranks are fixed at dusk, and the couriers do not learn the porter's habits from one night to the next. Each courier plans as if alone and revises on what it learns; the team's policy is not optimised jointly, and against the frequencies the team planned on them is a reference, not a bound. The porter walks at two when both couriers are in their wards, so that no robot meets him. A laser passing a lobby's mouth gets a few returns off the door, fewer than the rule needs, and the domain treats the door as unseen there. The nights are cycled, not sampled, and the comparison is of rankings on the same nights and starts. One floor, two couriers, simulation only.

References

[1] C. E. Alchourrón, P. Gärdenfors, D. Makinson. On the logic of theory change: partial meet contraction and revision functions. *Journal of Symbolic Logic* 50(2):510–530, 1985.

[2] G. Aucher. A combined system for update logic and belief revision. *PRIMA 2004*, LNCS 3371, 1–17, 2005.

[3] A. Baltag, L. S. Moss, S. Solecki. The logic of public announcements, common knowledge, and private suspicions. *TARK*, 1998.

[4] A. Baltag, S. Smets. A qualitative theory of dynamic interactive belief revision. *Logic and the Foundations of Game and Decision Theory (LOFT 7)*, 11–58, 2008.

[5] T. Bolander, M. B. Andersen. Epistemic planning for single- and multi-agent systems. *Journal of Applied Non-Classical Logics* 21(1):9–34, 2011.

[6] M. Dalal. Investigations into a theory of knowledge base revision. *AAAI*, 475–479, 1988.

[7] H. van Ditmarsch, W. van der Hoek, B. Kooi. *Dynamic Epistemic Logic*. Springer, 2007.

[8] H. van Ditmarsch. Prolegomena to dynamic logic for belief revision. *Synthese* 147(2):229–275, 2005.

[9] M. Goldszmidt, J. Pearl. Qualitative probabilities for default reasoning, belief revision, and causal modeling. *Artificial Intelligence* 84(1–2):57–112, 1996.

[10] A. Grove. Two modellings for theory change. *Journal of Philosophical Logic* 17(2):157–170, 1988.

[11] H. Katsuno, A. O. Mendelzon. Propositional knowledge base revision and minimal change. *Artificial Intelligence* 52(3):263–294, 1991.

[12] F. Martín, J. Ginés Clavero, V. Matellán, F. J. Rodríguez. PlanSys2: a planning system framework for ROS2. *IROS*, 2021.

[13] W. Spohn. Ordinal conditional functions: a dynamic theory of epistemic states. In W. L. Harper, B. Skyrms (eds.), *Causation in Decision, Belief Change, and Statistics*, vol. II, 105–134. Kluwer, 1988.

[14] J. van Benthem. Dynamic logic for belief revision. *Journal of Applied Non-Classical Logics* 17(2):129–155, 2007.

[15] Open Robotics. Hospital models, Gazebo Fuel. `https://app.gazebosim.org/OpenRobotics`, CC BY 4.0.

[16] AWS RoboMaker. Small House World. `https://github.com/aws-robotics/aws-robomaker-small-house-world`, MIT-0.